TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	75.9 ± 3.30
Correlation Stability	90.3 ± 1.10
Distribution Consistency	86.0 ± 3.03
Entropy Preservation	48.1 ± 1.04
Mean Z Score Drift	0.3 ± 0.03
Categorical Fidelity	65.3 ± 3.07
Privacy Risk	2.7 ± 1.00
Data Density	100.0 ± 0.00
Coverage Score	59.3 ± 4.15
Feature Accuracy	69.2 ± 1.47
Novelty Score	0.9 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	google/gemma-4-e2b
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	40.485	-4.261	9.5%	OK
TB	3.299	1.698	-1.601	48.5%	Drift
DB	1.486	0.489	-0.997	67.1%	Drift
Alkphos	290.576	191.112	-99.465	34.2%	Drift
Sgpt	80.714	32.845	-47.868	59.3%	Drift
Sgot	109.911	39.549	-70.361	64.0%	Drift
TP	6.483	6.080	-0.403	6.2%	OK
ALB	3.142	2.881	-0.261	8.3%	OK
A/G Ratio	0.947	0.914	-0.033	3.5%	OK
Selector	1.286	1.103	-0.183	14.3%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	75.9 ± 3.30
Correlation Stability (*2)	90.3 ± 1.10
Distribution Consistency (*3)	86.0 ± 3.03
Entropy Preservation (*4)	48.1 ± 1.04
Mean Z Score Drift (*5)	0.3 ± 0.03
Categorical Fidelity (*6)	65.3 ± 3.07
Privacy Risk (*7)	2.7 ± 1.00
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	59.3 ± 4.15

Feature Accuracy (*10)	69.2 ± 1.47
Novelty Score (*11)	0.9 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.7119658119658119	0.6871794871794872	0.01282051282051282	
F1	0.7821343746683586	0.8284598600934243	0.8044941341143298	0.02235975944597124	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7145299145299145	0.7059829059829059	0.7205128205128205	0.00598290598290598	
F1	0.8154972140649285	0.8261528029988279	0.8294180643531904	0.01392085028826192	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6854700854700855	0.7068376068376068	0.7145299145299145	0.02905982905982907	
F1	0.7909408486522184	0.8255516940019823	0.8166991668215523	0.02575831816933883	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6897435897435897	0.6786324786324787	0.6982905982905983	0.00854700854700854	
F1	0.8011016273174306	0.8062418960323852	0.8171894932723027	0.01608786595487215	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.7076923076923076	0.7102564102564102	0.00427350427350428	
F1	0.8128002697439015	0.8277001433368552	0.8249167720121052	0.01211650226820385	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.7068376068376068	0.7068376068376068	0	
F1	0.8279091624529537	0.8279091624529537	0.8279091624529537	0	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.7128205128205127	0.7205128205128205	0.01794871794871795	
F1	0.7997903571549948	0.8292984176412984	0.8163404652574296	0.01655010810243484	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5589743589743594	0.6803418803418804	0.5923076923076923	0.03333333333333334	*
F1	0.5587432119380917	0.7725278294746187	0.6138910848456949	0.05514787290760312	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6641025641025642	0.6991452991452991	0.6854700854700855	0.02136752136752134	
F1	0.7663032365208943	0.8116224397898806	0.7896951512462479	0.02339191472535362	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.7068376068376068	0.7128205128205127	0.02649572649572648	
F1	0.7971299319768493	0.8279091624529537	0.8277803606086902	0.03065042863184074	*

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	56	Male	1.4506897941286678	0.1108741715398966	196	22	44	5.110755748339753	2.058077028671483	0.5485075422861231
2	26	Male	0.8665887466008302	0.041636460116106	197	40	28	5.83215970463974	2.2362077627226298	0.5617037230978511
3	63	Male	2.199160564206201	0.2965357867191053 5	243	50	68	6.0190944733566	2.1881816098151243	0.5047841605779495
4	55	Male	1.3162552044039368	0.5337313630045202	212	45	50	6.151694447615361	2.4336941986834253	0.7559751465539207
5	45	Male	0.4674079372121127	0.3752760380881729	119	37	68	7.2503089273952954	2.7940050707995683	1.0291354536058621
6	59	Male	1.0469717811182177	0.1408743688831189	198	14	27	5.720097643863168	1.65504621082456	0.6901636054475402
7	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
8	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
9	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
10	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	19	Male	1.6469294293583512	0.4418678194655075 7	146	39	53	7.1209781322194505	4.7384414740536105	0.8466622141471339
2	48	Male	1.4051489591810902	-0.12717999847089378	167	46	70	6.615496757222395	2.5094500609114774	0.629015200651 99
3	50	Male	1.2447150340072 96	0.2348626337764725	83	19	35	7.1909924328426 53	3.2266737455370587	0.8299039990246712
4	24	Male	1.5132635033637 91	1.9329279701429 17	227	35	85	7.0082684719565 04	2.0926745711074353	0.7942328936421729
5	50	Male	0.5110537742034571	0.0159882064842642 45	149	6	36	7.1686988668029 85	3.0384218063784 32	0.9379091645731363
6	24	Male	4.7954029244335334	1.6046172379609256	364	39	61	5.7724518252684 17	2.7876057092375497	0.4703048725547231
7	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
8	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
9	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
10	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	28	Male	1.3275899916928258	0.5193837718526113	268	27	64	6.852589152290659	3.886673240704443	1.5108185349023242
2	43	Male	1.0746925455259855	0.1358290536801757	195	27	31	4.833536076171194	2.350838063344941	0.5015067133707504
3	29	Male	0.4797320336291026	0.2627732742954605	173	29	47	7.556805239475309	3.387646091454658	1.0840389975249731
4	40	Male	1.0655010874132336	0.3623704982979148	242	42	58	4.319916000407378	2.694184308731090	0.5366528510599662
5	32	Male	0.5810638597790398	0.5165848511589283	111	16	-1	6.529312670939582	3.625542578568027	1.3297854045194732
6	15	Male	1.3099482128359732	0.4190484109422857	207	43	89	6.398223506669456	3.203776387309416	0.7044047586308938
7	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
8	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
9	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
10	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	54	Male	0.6702674013254186	-0.12026902219402147	187	15	47	6.64465742908024	3.762414324493943	1.5891804506638991
2	62	Male	1.2399211464698965	0.4652850515290746	144	50	58	6.980566921618799	4.016755473890894	1.6671733115073868
3	62	Male	0.7197747533936727	0.1981132429415333	100	24	-6	6.772670692456935	2.921336689559116	1.0855734534898125
4	57	Male	1.5607404379859868	0.8467071343268722	254	25	36	5.983154832387858	1.775274849567191	0.4130564035853005
5	32	Male	1.1610964686600285	0.2646798454688567	207	32	-13	6.922020860777811	4.006979878240124	1.3148195395143947
6	47	Male	1.69072670369656	0.12548870995095915	194	72	45	4.97942213575236	2.37154070628235	0.9542036043968065
7	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
8	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
9	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
10	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	59	Male	1.2357250845607748	0.05767717518621884	188	38	51	4.788621463793376	1.3959985286257477	0.7188972718973676
2	37	Male	0.7665175244826992	0.32142770254649394	221	56	48	6.592936390265062	3.4761678935512257	1.0351440080378487
3	40	Male	1.1386707000155007	0.15057284753181377	192	1	23	7.868803874057857	3.1364895246995506	1.0830529631762666
4	42	Male	1.138655319090317	0.23955891298949258	237	41	25	6.600761739920617	4.0992439407151819	0.9491774866468478
5	28	Male	0.4495902665331043	0.4489205599985142	298	22	43	7.310068510874069	3.8904604265613356	0.7122334954915625
6	57	Male	1.6410843966167288	0.44520718114422125	148	25	70	6.661047808459481	3.8070185898397058	1.0090253892818477
7	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
8	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
9	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
10	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	49	Male	1.9965804491577435	0.5268598247632146	196	39	51	5.890716373479409	2.4405890831278727	0.6192118978601282
2	24	Male	1.529834364932909	0.7470208624518851	204	39	55	6.874214286648106	2.339084751917983	0.8295551476025286
3	18	Male	1.2059749178881227	0.4174422717862042	190	40	34	7.347323440777214	3.980614485757472	1.5739163807132732
4	50	Male	1.0845642089290806	0.16599067204484333	171	30	84	7.05812709099367	3.8809256503356018	0.7480599819174588
5	30	Male	1.039225361707167	0.23353432343772096	156	46	39	7.400278452112934	3.969998198574118	0.8158658998266893
6	32	Male	1.994098129901679	-0.12812873721688	203	33	26	7.766556479447131	4.558591977172602	1.3407574132544753
7	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
8	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
9	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
10	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	22	Male	4.48501561230974	5.55448846909626	238	38	21	7.48110336974672	3.56284619297685	1.26012254620798
			7					3	27	75
2	60	Male	0.84710148303951	0.52068669937651	179	17	48	6.65719483053971	3.35759734534744	1.05155225193018
			36	93				7	55	23
3	39	Male	0.63855435423805	0.08799201350556	157	41	14	6.93169470958605	3.85381658437100	1.12338207748307
			05	543				95	73	43
4	57	Male	1.59619809647664	0.22813393246126	123	43	30	8.27014362480048	4.60470879976430	1.82499268732720
			51	404				6	95	67
5	15	Male	0.57385049884200	0.24212269953575	198	38	27	7.57180621149029	3.88249211486228	1.16064652217575
			41	463				3	2	76
6	61	Male	1.39023634194124	0.16577467639754	178	38	49	6.95336384442889	3.95062115265753	1.02700883124626
			16	223				4	46	09
7	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
8	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
9	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
10	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	35	Male	0.40171822252324	1.30219375422194	191	39	35	6.14533464373506	2.22433834511117	0.45190798704442
			485	7					33	114
2	39	Male	0.59512622450700	0.33597915954606	162	32	54	6.30591221777656	2.71863906579069	0.88783624216963
			4	195				4	5	91
3	50	Male	0.72090167858130	0.22495267358768	191	22	88	6.12958741664575	2.25332832713226	0.79910800926734
			76	503					95	49
4	31	Male	0.92363019120146	0.27327481489458	185	23	51	6.36540279726131	2.89384513365784	0.68560438188450
			4	416				2	7	48
5	38	Male	0.95676309108002	0.20224003949994	182	19	55	6.15981742699116	4.47535602635023	1.00456028431394
			53	232				9	5	32
6	32	Male	1.02411790005406	0.06290685974163	124	41	66	7.09827948037387	2.90515522143156	0.61933077094852
			79	177				65	35	22
7	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
8	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
9	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
10	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	59	Male	1.37924882824977 12	0.44397262393930 73	140	41	-5	6.75086923566183 6	2.86125708024047 04	0.63833014654514
2	43	Male	1.30252808999793 75	0.54924058074696 76	213	33	40	5.43318467445168 7	3.65640960145333 16	0.84819863026900 79
3	53	Male	0.90027022905043 79	0.20243838994470 26	214	47	25	7.08069534180603 6	2.60963967137480 25	0.74778753959236 12
4	42	Male	2.10125369340893 7	1.36799336468334 95	172	10	55	5.55336230433959 7	3.28921917193016 2	0.63649888182930 17
5	36	Male	1.01491607469589 14	0.05315372143074 1124	127	28	39	8.09472953851758 3	4.09139770792328 4	1.68437547356704 64
6	34	Male	0.88111836024229 01	-0.06839085667105 405	195	17	35	7.61837316895061 44	3.11071207027732 97	0.84352118439759 86
7	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
8	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
9	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
10	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Male	1.28931306298192 58	-0.06833697603225 114	93	30	54	7.20624866515431 6	3.99029709483904 86	1.08827862479737 66
2	32	Male	1.27355916069468 83	0.19645657482909 387	246	22	50	6.64899528972395 4	2.79023739372603 1	1.13670077637043 95
3	42	Male	1.01070603584765 8	0.33108699312869 12	148	34	26	6.02876158537397 4	3.83296446812809 77	1.55793546537220 37
4	31	Male	22.1655178880708 73	12.5136558296350 29	247	63	34	7.12915430560995 76	2.83847337308939 3	0.66351354172000 89
5	37	Male	0.83976902771736 45	0.01628057767726 8093	150	45	14	7.26207678114932 7	4.18229890766280 7	0.89818156145650 96
6	63	Male	11.7104326264914 71	2.04812843746806 2	195	20	60	5.11937617754813 4	2.50828267041674 56	0.73772899490661 97
7	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
8	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
9	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
10	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3